

Enterprise Learning Platform

with Skill and Career Guidance System

Milestone 3 — Project Documentation

Engagement, Career Tools & Community

Team C

Prepared for team review

1. Overview — Where We Resumed From

Milestone 2 turned the platform into a working learning system: a real dashboard, a course catalog with enrollment, and real lesson content with server-graded quizzes. Milestone 3 broadens the platform from “a place to take a course” into a complete student experience — adding assessment tools, gamification, community features, and the career-facing tools that give the platform its “Skill and Career Guidance” identity, not just its learning-management identity.

1.1 What Was Already Done (Milestones 1–2 Recap)

- Google OAuth2 + JWT authentication, email/password login, forgot/reset password via email
- Real dashboard with live stats, streaks, weekly activity chart, and dynamic achievements
- Course catalog with real enrollment, and a lesson player supporting video, reading, and server-graded quiz lessons
- Account/profile settings page with editable personal details

2. Milestone 3 — Feature Summary

2.1 Assignments & Practice

Beyond in-lesson quizzes, students can now submit and track dedicated Assignments per course, and use a separate Practice section for extra exercises outside the graded course flow — giving students a lower-stakes way to reinforce a skill before or after a graded assessment.

2.2 Certificates & Verification

On completing a course, students earn a digital Certificate with a unique verification number. A Certificates Hub lets students view and download everything they've earned, and a public Certificate Verification page lets anyone — an employer, for instance — confirm a certificate is genuine without needing to log in.

2.3 Resume Builder

A built-in Resume Builder lets students assemble a resume directly from the platform, using multiple templates and pulling in real course/certificate achievements — directly supporting the platform's career-guidance goal rather than treating learning and job-readiness as separate concerns.

2.4 Internships, Live Sessions & Schedule

- An Internships listing surfaces opportunities relevant to a student's learning path.
- Live Sessions support scheduled, instructor-led classes beyond self-paced lessons.
- A personal Schedule page gives each student a single place to see upcoming sessions and deadlines.

2.5 Community: Forum, Leaderboard & Gamification

- Forum — students can post threads, reply, and upvote helpful answers, giving the platform a peer-support layer beyond one-directional course content.
- Leaderboard — ranks top learners platform-wide, building on the achievement system introduced in Milestone 2.
- Expanded badges — gamification achievements now cover a wider range of behaviors, not just streaks and course completion.
- Notifications & Announcements — course-level announcements and personal notifications keep students informed without needing to check every page manually.

2.6 Analytics & Settings

A personal Analytics page gives students a deeper view into their own learning statistics beyond the dashboard summary, and the account area was split so general app preferences live in a dedicated Settings page separate from the Profile page's personal-details editing.

3. Database Additions

Entity	Purpose
Assignment / AssignmentSubmission	Assignment tasks and student submissions
Badge / UserBadge	Available badges and which users have earned them
Certificate	Issued certificates, verifiable by certificate number
ForumThread / ForumReply / ForumUpvote	Discussion forum data
Internship	Internship listings
LiveSession	Scheduled live class sessions
Announcement / Notification	Course announcements and user notifications
Resume	User's resume builder data

As in prior milestones, all new tables were created automatically by Hibernate — no manual SQL was required, and no existing data was affected.

4. New Backend Controllers

Controller	Responsibility
AssignmentController	Assignment submissions
BadgeController / LeaderboardController	Gamification: badges & rankings
CertificateController	Issue & verify certificates
ForumController	Discussion threads, replies, upvotes

Controller	Responsibility
InternshipController	Internship listings
LiveSessionController / ScheduleController	Live classes & personal schedule
ResumeController	Resume builder data
NotificationController / AnnouncementController	Notifications & announcements

5. New Frontend Pages

File	Purpose
Assignments.jsx	Submit and track assignment work
Quizzes.jsx	Standalone quiz view and scored results
Practice.jsx	Extra practice exercises
CertificatesHub.jsx	View and download earned certificates
VerifyCertificate.jsx	Public certificate verification page
ResumeBuilder.jsx	Build and export a resume
Internships.jsx	Browse internship listings
Sessions.jsx / Schedule.jsx	Live sessions and personal schedule
Forum.jsx	Post threads, reply, and upvote
Leaderboard.jsx	Top-ranked learners
Analytics.jsx	Personal learning statistics
Settings.jsx	App preferences, separate from Profile

6. Points Worth Confirming With the Team

- The “saved course” wishlist heart icon currently stores its data in the browser's local storage rather than the backend, so saved courses do not sync across devices — worth confirming whether that's intentional for now.
- Forum moderation (editing/removing inappropriate posts) is not yet built.
- Certificate design/branding should be finalized before certificates are shared externally by students.

7. Current Project Status

Area	Status
Assignments & Practice	Complete & verified
Certificates & public verification	Complete & verified
Resume Builder	Complete & verified
Internships listing	Complete & verified
Live Sessions & Schedule	Complete & verified
Forum (threads, replies, upvotes)	Complete & verified
Leaderboard & expanded badges	Complete & verified
Notifications & Announcements	Complete & verified
Analytics & Settings pages	Complete & verified
Wishlist synced to backend (vs. local storage)	Not yet done

8. Recommended Next Steps

- Move the wishlist feature from local storage to a proper backend-stored preference for cross-device sync.
- Add forum moderation tools for instructors/admins.
- Layer AI-powered guidance (chatbot, personalized roadmap, skill-gap analysis) on top of this feature set — planned for Milestone 4.
- Build out Instructor and Admin tooling to manage the content and community introduced in this milestone — also planned for Milestone 4.